TMM TESTS

1. Which of the following is a key characteristic of ferrous alloys?

A) High corrosion resistance
B) High tensile strength
C) Primarily composed of iron
D) Non-magnetic

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2. In the Bessemer process, which of the following is used to remove impurities from molten iron?

A) Oxygen
B) Nitrogen
C) Carbon dioxide
D) Hydrogen

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3. The Hall-Héroult process is used for the extraction of which metal?

A) Iron
B) Aluminum
C) Copper
D) Zinc

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4. Which of the following is an example of a non-ferrous alloy?

A) Bronze
B) Steel
C) Cast iron
D) Stainless steel

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5. What is the purpose of adding carbon to iron in the production of steel?

A) To increase ductility
B) To improve corrosion resistance
C) To enhance hardness and strength
D) To make it more malleable

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6. Which phase in the iron-carbon diagram is stable at room temperature for most carbon steels?

A) Austenite
B) Ferrite
C) Cementite
D) Pearlite

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7. What is the main function of flux in the extraction of metals?

A) To reduce the metal
B) To purify the metal
C) To increase the melting point
D) To lower the temperature

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8. Which of the following is an example of a refractory metal?

A) Lead
B) Molybdenum
C) Copper
D) Gold

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9. Which of the following methods is used for extracting zinc from its ores?

A) Pyrometallurgy
B) Hydrometallurgy
C) Electrolytic refining
D) All of the above

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10. The critical point in the iron-carbon diagram, where austenite transforms into pearlite, is known as:

A) Eutectoid point
B) Eutectic point
C) Peritectic point
D) Solidus point

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11. In the production of stainless steel, which element is added to improve corrosion resistance?

A) Chromium
B) Nickel
C) Manganese
D) Silicon

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12. Which of the following elements is considered a heavy metal in metallurgy?

A) Lithium
B) Lead
C) Aluminum
D) Magnesium

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13. What is the primary purpose of carburizing in heat treatment of steel?

A) To increase hardness by adding carbon
B) To improve toughness
C) To soften the material
D) To increase ductility

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14. What is the most common method for refining copper?

A) Electrolytic refining
B) Smelting
C) Roasting
D) Carbon reduction

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15. What is the eutectic composition of the iron-carbon alloy system?

A) 0.8% C
B) 2.0% C
C) 0.5% C
D) 4.3% C

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16. Which of the following is NOT a method of refining metals?

A) Distillation
B) Electrolytic refining
C) Zone refining
D) Calcination

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17. What is the main application of titanium alloys in metallurgy?

A) High-temperature applications
B) Electrical conductors
C) Magnetic materials
D) Corrosion-resistant coatings

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18. Which of the following alloys is mainly used for making electrical conductors?

A) Brass
B) Bronze
C) Copper
D) Steel

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19. Which process is used to extract iron from its ore in the blast furnace?

A) Reduction
B) Oxidation
C) Electrolysis
D) Roasting

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20. In powder metallurgy, which of the following processes is used for shaping metal powders into a desired shape?

A) Sintering
B) Casting
C) Forging
D) Welding

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21. Which of the following materials is used as a catalyst in the Haber process for ammonia synthesis?

A) Iron
B) Copper
C) Nickel
D) Molybdenum

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22. Which alloy is primarily used in the construction of aircraft engines?

A) Aluminum alloy
B) Brass
C) Cast iron
D) Stainless steel

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23. What is the effect of adding tungsten to steel?

A) Increased brittleness
B) Increased hardness at high temperatures
C) Improved ductility
D) Reduced tensile strength

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24. The process of removing sulfur from molten steel is known as:

A) Deoxidation
B) Desulfurization
C) Decarburization
D) Refining

25. What is the main drawback of using lead in alloys?

A) High cost
B) Environmental pollution
C) High density
D) Poor corrosion resistance

26. What is the primary characteristic of cast iron?

A) High tensile strength
B) High ductility
C) Brittle with high carbon content
D) Low melting point

27. In the production of steel, what is the purpose of the Basic Oxygen Steelmaking (BOS) process?

A) To reduce carbon content
B) To add chromium
C) To remove sulfur
D) To solidify molten iron

28. The phenomenon of formation of a solid solution between two metals is known as:

A) Alloying
B) Phase separation
C) Eutectoid reaction
D) Diffusion

29. Which of the following materials is most commonly used for the production of superconducting materials?

A) Gold
B) Copper
C) Niobium
D) Iron

30. The process of cooling molten metal in a mold to form a solid shape is known as:

A) Forging
B) Casting
C) Rolling
D) Extrusion
